AUDITORY GROUPING DAILY

Tuesday, 07 July 2026 · 269 papers screened · curated

Today's strongest thread is about boundaries: an eLife study formally dissociates prediction-error-driven from prediction-uncertainty-driven event-segmentation signals in different cortical networks, a bioRxiv birdsong paper shows striatal dopamine flagging sequence onsets specifically rather than sequence content throughout, and a Current Biology study shows a relational/structural map in medial prefrontal cortex only crystallizing after days of consolidation - three different angles on the same question underlying your own position-invariant target-generalization problem. On the modelling side, a TMS study on urgency regulation and a CPP evidence-accumulation study both offer direct, testable extensions to the urgency and drift components of your leaky-accumulator DDM. It's a quieter day for core auditory-cortex physiology specifically - the closest auditory-adjacent hits are a somatosensory MMN/SPN analogue and gerbil auditory-nerve tinnitus data rather than anything in auditory cortex or prelimbic cortex directly, which is worth flagging honestly rather than padding around.

CHUNKING & SEQUENCE PROCESSING

Striatal Dopamine at Learned Sequence Boundaries Sustains Birdsong

BIORXIV *Xiao, L. et al. — bioRxiv — 2026-07-06*

Phasic striatal dopamine transients occur specifically at the initiation of a learned song sequence (not throughout it), shift progressively earlier - toward sequence onset - as the song matures, and are causally required (via temporally-targeted optogenetic inhibition) for long-term maintenance of the learned sequence. This is a direct precedent for a boundary/onset-locked dopaminergic teaching signal that could interact with your GLM's Module C (slow satiety/time-on-task drift) or with reward-locked events in the go/no-go task, and raises the question of whether the onset of your BC target sequence specifically - as opposed to its constituent chords - carries privileged reward-prediction weight.

An abstract relational map emerges in the human medial prefrontal cortex with consolidation.

PUBMED *Alon B Baram et al. — Current Biology — 2026 — Oxford Centre for Integrative Neuroimaging, FMRIB, Nuffield… — Germany*

Using fMRI repetition suppression and RSA, this study finds an abstract relational map of learned graph structure emerging in human medial prefrontal cortex only after several days of consolidation between two scanning sessions, not immediately after learning. That gives a concrete, testable consolidation-timescale prediction for your own prelimbic recordings: if PL is going to show a genuinely position-invariant representation of the BC target (the still-unsolved BC = XBC = XXBC generalization problem), this result suggests that abstraction may only emerge after extended training rather than being present from the first sessions.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Emergence of behavioral tinnitus in gerbils is associated with reduced spontaneous rates in single auditory nerve fibers.

PUBMED *Amarins N Heeringa — Journal of Neuroscience — 2026 — Research Centre Neurosensory Science, Department of Neuroscience,… — Germany*

Single-unit recordings from gerbil auditory nerve fibers show that behaviorally-confirmed tinnitus is associated with a reduction (not increase) in spontaneous firing rate, specific to fibers tuned near the affected frequency band - challenging the simple view that central auditory hyperactivity in tinnitus just reflects peripheral hyperactivity. This is a useful floor-effect consideration when interpreting frequency-specific spontaneous rates in your own auditory-nerve/auditory-cortex single-unit yield, given that your C57BL/6 mice have their own well-characterized progressive high-frequency hearing loss that could similarly confound baseline firing rates with genuine sequence-related changes.

BEHAVIORAL & COMPUTATIONAL MODELING

Motor implementation of context- and reward-based urgency regulation during action-oriented decision making.

PUBMED *Thibault Fumery et al. — Journal of Neuroscience — 2026 — Institute of Neuroscience, CoActions Lab, Université catholique… — Belgium*

Using TMS-measured corticospinal excitability, this study dissociates context-based from reward-based sources of decision urgency and finds they have distinct motor signatures (broad excitability change vs. surround inhibition) that map onto trait impulsivity differently across individuals. This speaks directly to the time-dependent urgency/collapsing-bounds toggle in your leaky-accumulator DDM family - it suggests that a single collapsing-bound term may be conflating two mechanistically separable sources of urgency (task context vs. reward prospect) that could in principle be fit as separate parameters in your model comparison.

A Common Neural Signal of Evidence Accumulation for Perceptual and Mnemonic Decisions

BIORXIV *Tsvinev, A. et al. — bioRxiv — 2026-07-06*

This EEG study shows the centro-parietal positivity (CPP) - already established as an evidence-accumulation signature in perceptual 2AFC tasks - also tracks accumulation slope in semantic and episodic memory-based decisions, arguing for a domain-general accumulation signal rather than one specific to sensory evidence. That is relevant to whether a shared accumulation signature might link chord-by-chord sensory evidence in your go/no-go task to position-independent template recognition, which bears on your open question of generalizing the target response across different preceding-chord contexts.

STATISTICAL LEARNING & ODDBALL

Learning regularities in noise engages both neural predictive activity and representational changes

RSS:NATURE COMMUNICATIONS *Nature Communications — 2026-07-07*

Using MEG and multivariate pattern analysis, this study dissociates two temporally distinct mechanisms for extracting regularities from noise: a rapid predictive-activity response, followed later by a slower change in the underlying stimulus representation itself. This two-stage account is directly relevant to interpreting your mouse behavioral data - it offers a candidate explanation for why distractor confusions might reflect a fast, expectation-like signal (closer to your GLM's trial-history Module B) that is dissociable from a slower-forming, genuinely learned chunk-identity representation (Module A).

Does stimulus preceding negativity reflect predictions in a somatosensory roving paradigm?

PUBMED *Gianluigi Giannini et al. — Journal of Neuroscience — 2026 — Neurocomputation and Neuroimaging Unit (NNU), Freie Universität… — Germany*

In a somatosensory roving (oddball) paradigm, this study identifies a stimulus-preceding negativity (SPN) - localized to inferior frontal, fronto-polar, and somatosensory cortex - that scales parametrically with the length of the preceding run of repeated stimuli, i.e., a genuinely anticipatory, pre-stimulus signature of accumulated stimulus history. This is a close conceptual analogue to what you might look for in prelimbic cortex just before a target chord: a prefrontal signal that reflects accumulated evidence from preceding chords rather than only the post-chord discrimination response captured by your current decision-window SDT analysis.

CIRCUITS, METHODS & POPULATION ANALYSIS

Complementary roles of cell-type-specific plasticity in shaping neocortical dynamics for learning action timing.

PUBMED *Shouvik Majumder et al. — Nature Communications — 2026 — Max Planck Florida Institute for Neuroscience, Jupiter, FL, USA.… — USA*

Using cell-type-specific CaMKII manipulations alongside large-scale electrophysiology, this study shows that plasticity in pyramidal-tract (but not intratelencephalic) premotor neurons is specifically required for learning action timing, and that these two projection classes make complementary contributions to population dynamics that anticipate motor timing. This is a useful circuit-dissociation precedent if you ever want to split your Neuropixels yield in prelimbic or auditory cortex by projection class when asking which population is doing the work of anticipating the timing of the target chord within your sequence.

Beyond DSA: Conjugacy-based Comparison of Dynamical Systems

ARXIV *Prakhar Godara et al. — arXiv (q-bio.NC) — 2026-07-05*

This methods paper identifies a mathematical gap in Dynamical Similarity Analysis (DSA), a popular method for testing whether two neural population trajectories implement the same underlying computation, and proposes a stricter conjugacy-based alternative that only permits alignments corresponding to genuine state-space bijections. Directly relevant if you use population-geometry or state-space comparisons (e.g., comparing prelimbic vs. auditory cortex trajectories, or comparing sessions/animals) - it flags that DSA-based claims of 'the same dynamics' may not hold up without this stricter criterion.

OTHER NOTABLE

Development of auditory and spontaneous movement responses to music over the first postnatal year

RSS:ELIFE *eLife — 2026-07-07*

Combining EEG auditory responses with markerless pose-estimation-based movement tracking across the first postnatal year, this study finds that auditory encoding of music emerges early in infancy while structured spontaneous movement to music only appears by 12 months. It's a nice cross-population parallel to your standing interest in treating spontaneous, unprompted movement as a first-class behavioral signal (your GLM's Module C framing) rather than trial-structure noise, and a demonstration of the same continuous pose-estimation approach applied to an auditory paradigm.

PAPER OF THE DAY

Multiple event segmentation mechanisms in the human brain

eLife — 2026-07-07 — new pick — abstract only

- The abstract lays out two explicit computational models that generate event boundaries from continuous experience by different rules - one triggered by prediction ERROR (an outright mismatch between expected and observed input) and one by prediction UNCERTAINTY (reduced confidence in the prediction, without necessarily being wrong) - which maps cleanly onto the mechanistic distinction your chunk-in-stream task's Start-Target/End-Target/Distractor trial types could in principle probe.
- Multivariate fMRI analysis around human-identified boundaries shows a specific three-stage temporal signature: an early pattern shift in anterior temporal regions around 11.9 seconds before the boundary, a later shift in parietal regions around 4.5 seconds before it, and whole-brain pattern stabilization about 11.8 seconds after - a concrete, reusable temporal template for where to look for a comparable signal in high-gamma or spike-timing data, though the absolute timescale (fMRI-slow) will not translate directly to your millisecond-resolution recordings.
- Critically, error-driven and uncertainty-driven boundaries are not just theoretically distinct but anatomically dissociable: error-driven boundaries are tied to early ventrolateral prefrontal pattern shifts followed by prefrontal/temporal stabilization, while uncertainty-driven boundaries instead recruit parietal regions within the dorsal attention network with comparatively little subsequent stabilization - meaning a single 'boundary detector' framing may be masking two separable circuits.
- This bears directly on your still-open question of whether committing to (or rejecting) a target sequence - e.g., the moment the mouse should discriminate BC from a distractor, or a human participant should recognize an embedded target chunk - reflects a genuine violation-of-expectation signal versus a drop in predictive confidence as evidence accumulates; these two accounts predict different prefrontal versus parietal-analog circuit involvement and different timing relative to the behavioral response.

HOW TO READ IT (≈20 min)

About 20 minutes, working from the abstract since the full text wasn't fetched (eLife article, likely too recently published to be indexed for full-text retrieval yet - worth revisiting via eLife's site directly if you want figures). Start with the framing of the two computational models (error-driven vs. uncertainty-driven boundary generation) - this is the conceptual core to hold onto. Then focus on the three-stage temporal sequence (anterior temporal -> parietal -> whole-brain stabilization) and the anatomical dissociation (ventrolateral prefrontal for error-driven vs. dorsal-attention/parietal for uncertainty-driven) - these are the two results most directly portable to your own recordings. If you pull the full text later, prioritize the Methods for exactly how 'prediction error' and 'prediction uncertainty' were operationalized in the computational models, since that operational definition is what you'd need to adapt to your own trial-type comparisons.

YOUR QUESTION TO ANSWER

In your human chunk-in-stream task, would you predict that correctly detecting a chunk boundary (the transition into a memorized target sequence) is signaled more by a prediction-error-like mechanism (surprise at a violation, expected to recruit prefrontal-then-temporal stabilization) or a prediction-uncertainty-like mechanism (reduced confidence as evidence accumulates, expected to recruit parietal/dorsal-attention regions) - and which of your existing trial types (Start-Target, End-Target, or Distractor) could already dissociate these two accounts in your recorded regions, versus requiring a new trial type that manipulates predictability without introducing an actual violation?

Source status — Flagship journals (Tier 1): 270 papers | bioRxiv deep scan (Tier 2): 10 papers | OpenAlex (Tier 3): 27 papers | PubMed (Tier 3): 10 papers | arXiv (Tier 3): 2 papers